# Hospital use among adults prescribed antidepressants in NHS Tayside

*A Fleming,[1] Dr. L. Ciaccio ,[1,2] Dr. C. Marwick, [1,3] Dr. K. Barnett [1,2]*

*[1] University of Dundee, [2] Department of Public Health, NHS Tayside, [3]University of St. Andrew's*

## Background

- People prescribed antidepressants represent a large, diverse and clinically heterogeneous population[1]
- Patterns of hospital use in this population remain poorly characterised

**Aim** Characterise hospital use among adults prescribed antidepressants by age, sex and socioeconomic deprivation and compare rates with the Scottish population

## Methods

*Retrospective observational study using anonymised linked data*

**1 Define the population | Jan 2019-Dec 2023**

NHS Tayside residents aged ≥16 with ≥1 antidepressant prescription in the corresponding month

Eligibility reassessed monthly

**2 Link prescribing and hospital records**

General acute admissions | A&E attendances | Psychiatric admissions

**3 Calculate monthly hospital use rates**

Calculate separately for each hospital setting

$$\frac{\text{Hospital contacts among adults prescribed antidepressants that month}}{\text{Number of adults prescribed antidepressants that month}} \times 1{,}000 = \text{Monthly hospital use rate}$$

*Contacts per 1,000 adults prescribed antidepressants*

Stratified by:

**Age** | 7 groups | **Sex** | Male / Female | **Deprivation** | SIMD quintiles

**Compare mean monthly rates:**

Sex: paired t-tests

Age / deprivation: one-way ANOVA + Tukey

$p < 0.05$

**4 Compare with Scottish population rates**

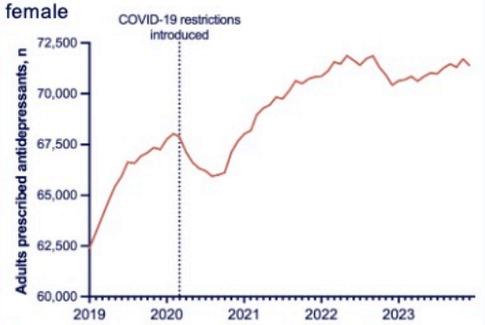
Compare age-specific and sex-specific population rates

$$\frac{\text{Tayside antidepressant-prescribed population rate}}{\text{Corresponding Scottish population rate}} = \text{Rate ratio}$$

## Results

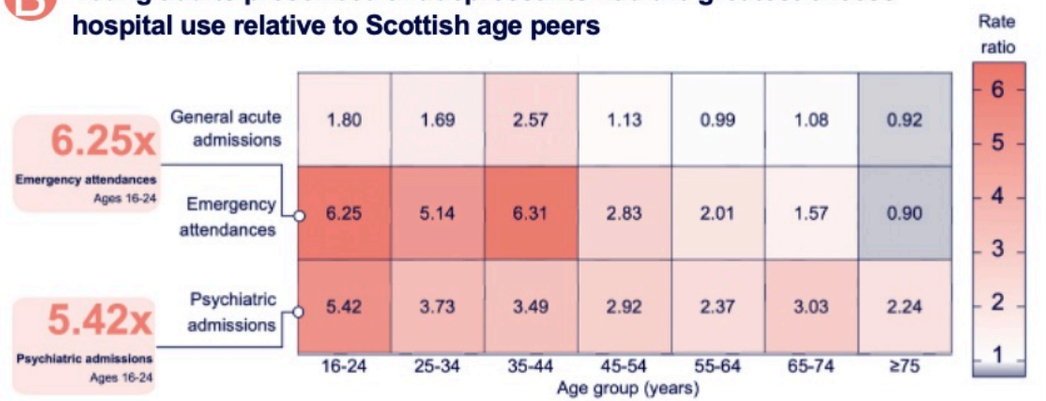
### A A large and growing population are prescribed antidepressants

Mean monthly population: 68,876 adults prescribed antidepressants (1 in 5 Tayside adults), 66.1% female




Prescribed population increased by 8.3% (Jan 2019-Dec 2023)

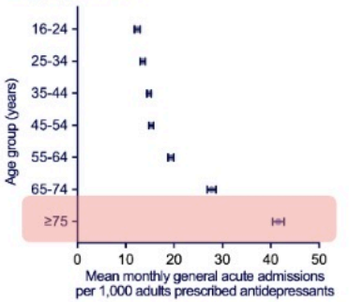
### B Young adults prescribed antidepressants had the greatest excess hospital use relative to Scottish age peers




Mean rate ratios, 2019-2023: Tayside adults prescribed antidepressants vs age-matched Scottish population

**Rate ratio = 1: equal rates | >1: higher use | <1: lower use**

### C Highest general acute rates at ≥75: Near parity with Scottish age peers




AGE ≥75:

**Absolute rate**

**41.86**

Admissions per 1,000 adults prescribed antidepressants/month

**Relative rate**

**0.92x**

The Scottish general acute admission rate for adults ≥75

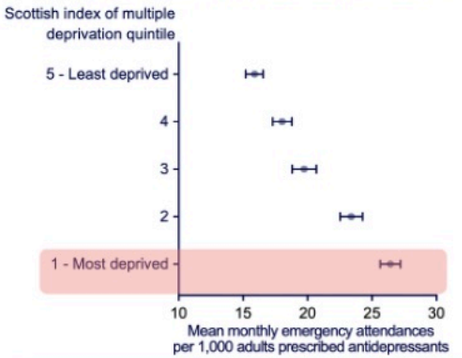
### D Men had greater excess hospital use




Mean rate ratios, 2019-2023: Tayside adults prescribed antidepressants Scottish population rates for the same sex. **1 = equal rates, >1 = higher use**

Across all three settings, men prescribed antidepressants had greater relative excess than women, and higher absolute rates of hospital use

### E Emergency attendance rates were higher with greater deprivation




**66% higher**

**Emergency attendance rate**

Most vs least deprived adults prescribed antidepressants

## Discussion and Conclusions

**Younger adults prescribed antidepressants had the greatest relative excess in hospital use**

Those aged ≥75 had the highest general acute and emergency admission rates, but little excess

**Next steps:**

- Invesigate drivers of excess hospital use among adults prescribed antidepressants, particularly young adults, alongside the deprivation gradient
- Use age, sex and deprivation patterns to guide service planning for adults prescribed antidepressants

**Explore the study**

QR CODE

**Reference:**

1. Brsnik V, Rottenkolber M, Vukas J, Schechner M, Lukaschek K, Jung-Sievers C, et al. Potential deprescribing indications for antidepressants between 2012 and 2019: repeated cross-sectional analysis in two Scottish health boards. BMC Med. 2024;22:378. doi:10.1186/s12916-024-03584-9